

**City and County of San Francisco
Office of Contract Administration
Purchasing Division
City Hall, Room 430
1 Dr. Carlton B. Goodlett Place
San Francisco, California 94102-4685**

**Agreement between the City and County of San Francisco and
United States Geological Survey**

**CS-1024
Technical Support for Alameda Creek Watershed Hydrological Model Development**

This Agreement is made this 15th day of October, 2015, in the City and County of San Francisco, State of California, by and between: **United States Geological Survey**, a bureau of the **United States Department of the Interior**, hereinafter referred to as "USGS," and the City and County of San Francisco, a municipal corporation, hereinafter referred to as "City," acting by and through its Director of the Office of Contract Administration or the Director's designated agent, hereinafter referred to as "Purchasing."

Recitals

WHEREAS, the San Francisco Public Utilities Commission ("Department") requires technical support for the development of a hydrological model for the Alameda Creek Watershed, using the USGS GSFLOW model; and,

WHEREAS, the USGS is authorized to perform technical assistance with other Federal agencies, units of State or local government, industrial organizations, private corporations, public and private foundations, and nonprofit organizations (including universities) under the Stevenson-Wydler Act (15 U.S.C. § 3710a, as amended); and,

WHEREAS, USGS represents and certifies that it is qualified to perform the services required by City as set forth under this Agreement; and,

WHEREAS, approval for this Agreement was obtained when the Civil Service Commission approved Contract number **34564 -15/16** on **August 19, 2015**;

Now, THEREFORE, the parties agree as follows:

1. Certification of Funds; Budget and Fiscal Provisions; Termination in the Event of Non-Appropriation. This Agreement is subject to the budget and fiscal provisions of the City's Charter. Charges will accrue only after prior written authorization certified by the Controller, and the amount of City's obligation hereunder shall not at any time exceed the amount certified for the purpose and period stated in such advance authorization. This Agreement will terminate without penalty, liability or expense of any kind to City at the end of any fiscal year if funds are not appropriated for the next succeeding fiscal year. If funds are appropriated for a portion of the fiscal year, this Agreement will terminate, without penalty, liability or expense of any kind at the

end of the term for which funds are appropriated. City shall promptly notify USGS if funds are not appropriated. City has no obligation to make appropriations for this Agreement in lieu of appropriations for new or other agreements. City budget decisions are subject to the discretion of the Mayor and the Board of Supervisors. USGS's assumption of risk of possible non-appropriation is part of the consideration for this Agreement. Pursuant to the Anti-Deficiency Act, 31 U.S.C. §1341 (a)(1), nothing herein contained shall be construed as binding the USGS to expend in any one fiscal year any sum in excess of its appropriations or funding in excess or what it has received for the collaborative work outlined in Appendix A.

THIS SECTION CONTROLS AGAINST ANY AND ALL OTHER PROVISIONS OF THIS AGREEMENT.

2. Term of the Agreement. Subject to Section 1, the term of this Agreement shall be from **November 1, 2015 to October 31, 2017**. The term will be November 1, 2015 to October 31, 2016 if terminated earlier.

3. Effective Date of Agreement. This Agreement shall become effective when the Controller has certified to the availability of funds and USGS has been notified in writing.

4. Services USGS Agrees to Perform. The USGS agrees to perform the services provided for in Appendix A, "Services to be provided by USGS," attached hereto and incorporated by reference as though fully set forth herein.

5. Compensation. Compensation shall be made in annual payments. The City shall make a one-time payment of **\$25,000 (twenty-five thousand dollars)** to USGS within 60 days of the Effective Date for the first year of services. Should the City authorize, in writing, the USGS to perform a second year of services, the City shall make a one-time payment of **\$25,000 (twenty-five thousand dollars)** to USGS within 60 days of this written authorization. In no event shall the amount of this Agreement exceed **Fifty Thousand Dollars (\$50,000)**. The breakdown of costs associated with this Agreement appears in Appendix B, "Calculation of Charges," attached hereto and incorporated by reference as though fully set forth herein. City may withhold payment to USGS in any instance in which USGS has failed or refused to satisfy any material obligation provided for under this Agreement. Invoices not paid within sixty (60) days of receipt bear interest at the annual rate established by the U.S. Treasury, pursuant to 31 USC §3717.

6. Guaranteed Maximum Costs. The City's obligation hereunder shall not at any time exceed the amount certified by the Controller for the purpose and period stated in such certification. Except as may be provided by laws governing emergency procedures, officers and employees of the City are not authorized to request, and the City is not required to reimburse the USGS for, Commodities or Services beyond the agreed upon contract scope unless the changed scope is authorized by amendment and approved as required by law. Officers and employees of the City are not authorized to offer or promise, nor is the City required to honor, any offered or promised additional funding in excess of the maximum amount of funding for which the contract is certified without certification of the additional amount by the Controller. The Controller is not authorized to make payments on any contract for which funds have not been certified as available in the budget or by supplemental appropriation.

7. Payment; Invoice Format. Invoices furnished by USGS under this Agreement must be in a form acceptable to the Controller, and must include a unique invoice number. All amounts paid by City to USGS shall be subject to audit by City to the extent allowed under Federal law.

Payment shall be made by City to USGS at the address specified in the section entitled "Notices to the Parties."

8. Reserved. (Submitting False Claims; Monetary Penalties)

9. Reserved. (Disallowance)

10. Taxes. USGS shall pay all applicable taxes levied in connection with this Agreement or the Services, including any self-employment, possessory interest or sales and use taxes.

11. Payment Does Not Imply Acceptance of Work. The granting of any payment by City, or the receipt thereof by USGS, shall in no way lessen the liability of USGS to replace unsatisfactory work, equipment, or materials, although the unsatisfactory character of such work, equipment or materials may not have been apparent or detected at the time such payment was made. Materials, equipment, components, or workmanship that do not conform to the requirements of this Agreement may be rejected by City and in such case must be replaced by USGS without delay.

12. Qualified Personnel. Work under this Agreement shall be performed only by competent personnel under the supervision of and in the employment of USGS. USGS will comply with City's reasonable requests regarding assignment of personnel to the extent allowed under Federal law, but all personnel must be supervised by USGS. USGS shall commit adequate resources to complete the project within the project schedule specified in this Agreement.

13. Responsibility for Equipment. City shall not be responsible for any damage to persons or property as a result of the use, misuse or failure of any equipment used by USGS, or by any of its employees, even though such equipment be furnished, rented or loaned to USGS by City.

14. Independent Party; Payment of Taxes and Other Expenses

a. **Independent Party.** USGS or any agent or employee of USGS shall be deemed at all times to be an independent party and is wholly responsible for the manner in which it performs the services and work requested by City under this Agreement. USGS or any agent or employee of USGS shall not have employee status with City, nor be entitled to participate in any plans, arrangements, or distributions by City pertaining to or in connection with any retirement, health or other benefits that City may offer its employees. USGS or any agent or employee of USGS is liable for the acts and omissions of itself, its employees and its agents. USGS shall be responsible for all obligations and payments, whether imposed by federal, state or local law, including, but not limited to, FICA, income tax withholdings, unemployment compensation, insurance, and other similar responsibilities related to USGS's performing services and work, or any agent or employee of USGS providing same. Nothing in this Agreement shall be construed as creating an employment or agency relationship between City and USGS or any agent or employee of USGS. Any terms in this Agreement referring to direction from City shall be construed as providing for direction as to policy and the result of USGS's work only, and not as to the means by which such a result is obtained. City does not retain the right to control the means or the method by which USGS performs work under this Agreement.

b. **Payment of Taxes and Other Expenses.** Should City, in its discretion, or a relevant taxing authority such as the Internal Revenue Service or the State Employment Development Division, or both, determine that USGS is an employee for purposes of collection of any employment taxes, the amounts payable under this Agreement shall be reduced by amounts equal to both the employee and employer portions of the tax due (and offsetting any

credits for amounts already paid by USGS which can be applied against this liability). City shall then forward those amounts to the relevant taxing authority. Should a relevant taxing authority determine a liability for past services performed by USGS for City, upon notification of such fact by City, USGS shall promptly remit such amount due or arrange with City to have the amount due withheld from future payments to USGS under this Agreement (again, offsetting any amounts already paid by USGS which can be applied as a credit against such liability). A determination of employment status pursuant to the preceding two paragraphs shall be solely for the purposes of the particular tax in question, and for all other purposes of this Agreement, USGS shall not be considered an employee of City. Notwithstanding the foregoing, should any court, arbitrator, or administrative authority determine that USGS is an employee for any other purpose, then USGS agrees to a reduction in City's financial liability so that City's total expenses under this Agreement are not greater than they would have been had the court, arbitrator, or administrative authority determined that USGS was not an employee.

15. Insurance. USGS and City agree that each party will maintain in force, throughout the term of this Agreement, a program of insurance. USGS is part of the Federal Government of the United States. It may not use appropriated funds, without express statutory authority, to purchase insurance. As a result, the Federal Government is essentially a self-insurer with respect to (1) loss of, or damage to, Federal Government property and (2) damage to persons or property caused by employee acts or omissions while acting within the scope of their employment in accordance with the Federal Tort Claims Act (codified at 28 USC 2671 et seq.). USGS warrants that it is self-insured for purposes of Worker's Compensation.

16. Indemnification. Each party agrees to be responsible for the activities, including the negligence of their employees or agents. Under Federal law in effect at the time of the signing of this agreement, USGS is precluded from indemnifying, holding harmless, and defending the Collaborator for the activities set out in this agreement. Federal law does provide, however, through the Federal Tort Claims Act (28 U.S.C. §2671 et seq.), a means of addressing claims for personal injuries or property damage caused by the negligent or wrongful act or omission of any employee of the United States while acting within the course of his or her employment. Consistent with this legal authority, the USGS shall be liable, to the extent allowed by the Federal Tort Claims Act, for claims for personal injuries or property damage resulting from the negligent acts or wrongful act or omission of any USGS employee while, acting within the scope of his employment arising out of this Agreement.

17. Reserved. (Incidental and Consequential Damages)

18. Liability of City. EXCEPT AS PROVIDED FOR IN SECTION 16 (INDEMNIFICATION), CITY'S PAYMENT OBLIGATIONS UNDER THIS AGREEMENT SHALL BE LIMITED TO THE PAYMENT OF THE COMPENSATION PROVIDED FOR IN SECTION 5 OF THIS AGREEMENT. NOTWITHSTANDING ANY OTHER PROVISION OF THIS AGREEMENT, IN NO EVENT SHALL EITHER PARTY BE LIABLE FOR ANY SPECIAL, CONSEQUENTIAL, INDIRECT OR INCIDENTAL DAMAGES, INCLUDING, BUT NOT LIMITED TO, LOST PROFITS, ARISING OUT OF OR IN CONNECTION WITH THIS AGREEMENT OR THE SERVICES PERFORMED IN CONNECTION WITH THIS AGREEMENT.

19. Reserved. (Liquidated Damages)

20. Default; Remedies

a. **Default.** USGS shall be in default if USGS: (a) fails to perform any covenant (including a lapse in insurance coverage); (b) files or is the subject of a petition for bankruptcy or insolvency; or, (c) has a court-ordered receiver or trustee appointed with respect to USGS's assets. Each of the following shall constitute an event of default ("Event of Default") under this Agreement.

b. **Remedies.** On and after any Event of Default, City shall have the right to exercise its legal and equitable remedies, including, without limitation, the right to terminate this Agreement or to seek specific performance of all or any part of this Agreement. All remedies provided for in this Agreement may be exercised individually or in combination with any other remedy available hereunder or under applicable laws, rules and regulations. The exercise of any remedy shall not preclude or in any way be deemed to waive any other remedy.

21. Termination for Convenience. Either party may terminate this Agreement without cause or penalty upon at least thirty days' prior written notice to the other party. In such event, USGS shall continue to perform the Services until the termination effective date, and USGS will be paid for those Services adequately performed through such date. Thereafter, this Agreement shall terminate, subject to Section 22. USGS disclaims any right to anticipated profits or losses or any other expense not expressly covered under Section 4 and incurred prior to such termination effective date.

22. Rights and Duties upon Termination or Expiration

This Section and the following Sections of this Agreement shall survive termination or expiration of this Agreement:

- | | |
|---|---|
| 8. Submitting false claims | 24. Proprietary or confidential information of City |
| 9. Disallowance | 26. Ownership of Results |
| 10. Taxes | 27. Works for Hire |
| 11. Payment does not imply acceptance of work | 28. Audit and Inspection of Records |
| 13. Responsibility for equipment | 48. Modification of Agreement. |
| 14. Independent USGS; Payment of Taxes and Other Expenses | 49. Administrative Remedy for Agreement Interpretation. |
| 15. Insurance | 50. Agreement Made in California; Venue |
| 16. Indemnification | 51. Construction |
| 17. Incidental and Consequential Damages | 52. Entire Agreement |
| 18. Liability of City | 56. Severability |
| | 57. Protection of private information |

Subject to the immediately preceding sentence, upon termination of this Agreement prior to expiration of the term specified in Section 2, this Agreement shall terminate and be of no further force or effect. USGS shall transfer title to City, and deliver in the manner, at the times, and to the extent, if any, directed by City, any work in progress, completed work, supplies, equipment, and other materials produced as a part of, or acquired with City funds in connection with the performance of this Agreement, and any completed or partially completed work which, if this Agreement had been completed, would have been required to be furnished to City. This subsection shall survive termination of this Agreement.

23. Conflict of Interest. Through its execution of this Agreement, USGS acknowledges that it is familiar with the provision of Section 15.103 of the City's Charter, Article III, Chapter 2 of City's Campaign and Governmental Conduct Code, and Section 87100 et seq. and Section 1090 et seq. of the Government Code of the State of California, and certifies that it does not know of any facts which constitutes a violation of said provisions and agrees that it will immediately notify the City if it becomes aware of any such fact during the term of this Agreement.

24. Proprietary or Confidential Information of City. USGS understands and agrees that, in the performance of the work or services under this Agreement or in contemplation thereof, USGS may have access to private or confidential information which may be owned or controlled by City and that such information may contain proprietary or confidential details, the disclosure of which to third parties may be damaging to City. USGS agrees that all confidential information disclosed by City to USGS shall be held in confidence and used only in performance of the Agreement. Any confidential or proprietary information disclosed by City to USGS shall be clearly marked, "Confidential." USGS shall exercise the same standard of care to protect such information as a reasonably prudent USGS would use to protect its own proprietary data.

25. Notices to the Parties. Unless otherwise indicated elsewhere in this Agreement, all written communications sent by the parties may be by U.S. mail, e-mail or by fax, and shall be addressed as follows:

To City: Amod Dhakal
San Francisco Public Utilities Commission
525 Golden Gate Avenue
San Francisco, CA 94102
adhakal@sfwater.org
(415) 554-0731; (925) 862-5515

To USGS: Dr. Rich Niswonger
USGS, NRP
345 Middlefield Road
Mail Stop 470
Menlo Park, CA 94025
rniswon@usgs.gov
(650) 329-4534; (775) 297-1392

Any notice of default must be sent by registered mail.

26. Ownership of Results. Any interest of USGS or its contractors s, in drawings, plans, specifications, blueprints, studies, reports, memoranda, computation sheets, computer files and media or other documents prepared by USGS or its contractors s in connection with services to be performed under this Agreement, shall become the property of USGS. However, City may retain and use copies for reference and as documentation of its experience and capabilities as long as they are not confidential or proprietary information of USGS or the Federal Government.

27. Reserved. (Works for Hire)

28. Audit and Inspection of Records. USGS shall maintain accurate books and records relating to this Agreement and the Services, including accounting records and copies of all invoices. USGS shall make such books and records available to the City for review and audit for

at least two years after termination of this Agreement, at a location that is readily accessible to the City. Any audit or inspection by the City shall be in accordance with Federal laws and regulations, and Department of Interior policies on access to Federal facilities, data and data systems.

29. Contracting. USGS is prohibited from contracting this Agreement or any part of it unless such contracting is first approved by City in writing. Neither party shall, on the basis of this Agreement, contract on behalf of or in the name of the other party. An agreement made in violation of this provision shall confer no rights on any party and shall be null and void.

30. Assignment. The services to be performed by USGS are personal in character and neither this Agreement nor any duties or obligations hereunder may be assigned or delegated by the USGS unless first approved by City by written instrument executed and approved in the same manner as this Agreement.

31. Non-Waiver of Rights. The omission by either party at any time to enforce any default or right reserved to it, or to require performance of any of the terms, covenants, or provisions hereof by the other party at the time designated, shall not be a waiver of any such default or right to which the party is entitled, nor shall it in any way affect the right of the party to enforce such provisions thereafter.

32. Reserved. (Consideration of Criminal History in Hiring and Employment Decisions) Waived because USGS is a public entity.

33. Reserved. (Local Business Enterprise Utilization; Liquidated Damages) Waived.

34. Reserved. (Nondiscrimination; Penalties) Deleted because USGS is a public entity.

35. Reserved. (MacBride Principles—Northern Ireland) Deleted because USGS is a public entity.

36. Reserved. (Tropical Hardwood and Virgin Redwood Ban)

37. Drug-Free Workplace Policy. USGS acknowledges that pursuant to the Federal Drug-Free Workplace Act of 1989, the unlawful manufacture, distribution, dispensation, possession, or use of a controlled substance is prohibited on City premises. USGS agrees that any violation of this prohibition by USGS, its employees, agents or assigns will be deemed a material breach of this Agreement.

38. Reserved. (Resource Conservation)

39. Compliance with Americans with Disabilities Act. USGS shall provide the Services in a manner that complies with the Americans with Disabilities Act (ADA), including but not limited to Title II's program access requirements, and all other applicable federal, state and local disability rights legislation.

40. Sunshine Ordinance. USGS acknowledges that this Agreement and all records related to its formation, USGS's performance of Services, and City's payment are subject to the California Public Records Act, (California Government Code §6250 et. seq.), and the San Francisco Sunshine Ordinance, (San Francisco Administrative Code Chapter 67). Such records are subject to public inspection and copying unless exempt from disclosure under federal, state or local law.

41. Reserved. (Public Access to Meetings and Records)

- 42. Reserved. (Limitations on Contributions)**
- 43. Reserved. (Requiring Minimum Compensation for Covered Employees) Deleted because USGS is a public entity.**
- 44. Reserved. (Requiring Health Benefits for Covered Employees) Deleted because USGS is a public entity.**
- 45. Reserved. (First Source Hiring Program) Deleted because USGS is a public entity.**
- 46. Prohibition on Political Activity with City Funds.** In accordance with San Francisco Administrative Code Chapter 12.G and to the extent required under Federal law, USGS may not participate in, support, or attempt to influence any political campaign for a candidate or for a ballot measure (collectively, "Political Activity") in the performance of the services provided under this Agreement. USGS agrees to comply with San Francisco Administrative Code Chapter 12.G and any implementing rules and regulations promulgated by the City's Controller. The terms and provisions of Chapter 12.G are incorporated herein by this reference. In the event USGS violates the provisions of this section, the City may, in addition to any other rights or remedies available hereunder, (i) terminate this Agreement, and (ii) prohibit USGS from bidding on or receiving any new City contract for a period of two (2) years. The Controller will not consider USGS's use of profit as a violation of this section.
- 47. Reserved. (Preservative-treated Wood Containing Arsenic)**
- 48. Modification of Agreement.** This Agreement may not be modified, nor may compliance with any of its terms be waived, except by written instrument executed and approved in the same manner as this Agreement.
- 49. Reserved. (Administrative Remedy for Agreement Interpretation)**
- 50. Agreement Made in California; Venue.** The formation, interpretation and performance of this Agreement shall be governed by Federal laws and the laws of the State of California. Venue for all litigation relative to the formation, interpretation and performance of this Agreement shall be in a court of competent jurisdiction in San Francisco.
- 51. Construction.** All paragraph captions are for reference only and shall not be considered in construing this Agreement.
- 52. Entire Agreement.** This contract sets forth the entire Agreement between the parties, and supersedes all other oral or written provisions. This contract may be modified only as provided in Section 48, "Modification of Agreement."
- 53. Compliance with Laws.** USGS shall keep itself fully informed of the City's Charter, codes, ordinances and regulations of the City and of all state, and federal laws in any manner affecting the performance of this Agreement, and must at all times comply with such applicable local codes, ordinances, and regulations and all applicable laws as they may be amended from time to time.
- 54. Services Provided by Attorneys.** Any services to be provided by a law firm or attorney must be reviewed and approved in writing in advance by the City Attorney. No invoices for services provided by law firms or attorneys, including, without limitation, as contractors of USGS, will be paid unless the provider received advance written approval from the City Attorney.

55. Reserved. (Supervision of Minors)

56. Severability. Should the application of any provision of this Agreement to any particular facts or circumstances be found by a court of competent jurisdiction to be invalid or unenforceable, then (a) the validity of other provisions of this Agreement shall not be affected or impaired thereby, and (b) such provision shall be enforced to the maximum extent possible so as to effect the intent of the parties and shall be reformed without further action by the parties to the extent necessary to make such provision valid and enforceable.

57. Protection of Private Information. The provisions of the San Francisco Administrative Code Chapter 12M, "Protection of Private Information," and any amendments thereto are incorporated herein by reference and made a part of this Agreement as though fully set forth to the extent allowed under Federal law.

58. Reserved.

59. Reserved. (Food Service Waste Reduction Requirements)

60. Reserved. (Slavery Era Disclosure)

61. Cooperative Drafting. This Agreement has been drafted through a cooperative effort of both parties, and both parties have had an opportunity to have the Agreement reviewed and revised by legal counsel. No party shall be considered the drafter of this Agreement, and no presumption or rule that an ambiguity shall be construed against the party drafting the clause shall apply to the interpretation or enforcement of this Agreement.

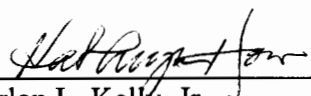
IN WITNESS WHEREOF, the parties hereto have executed this Agreement on the day first mentioned above.

CITY

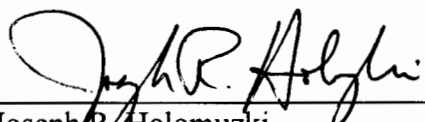
USGS

Recommended by:

United States Department of the Interior
Geological Survey



Harlan L. Kelly, Jr.
General Manager
San Francisco Public Utilities Commission

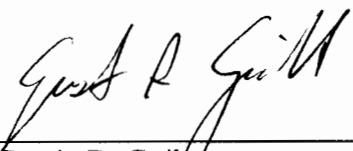
 10-16-15

Joseph R. Holomuzki,
Branch Chief

Approved as to Form:

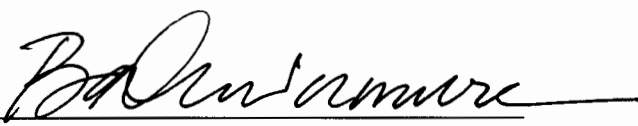
City vendor number: **98407**

Dennis J. Herrera
City Attorney

By: 

Gustin R. Guibert
Deputy City Attorney

Approved:



Jaci Fong
Director of the Office of Contract
Administration, and
Purchaser

Appendices

- A: Services to be provided by USGS
- B: Calculation of Charges

Appendix A

Services to be provided by USGS

1. Description of Services

The USGS shall provide technical support to the San Francisco Public Utilities Commission (SFPUC) for the development of a GSFLOW model for the Alameda Creek Watershed. Dr. Rich Niswonger from the U.S. Geological Survey (USGS) National Research Program—Western Branch will be the Primary Investigator (PI) on the agreement. The USGS PI will provide advice and technical support on these tasks. Model development will be performed solely by the SFPUC.

The watershed model development will be performed by SFPUC over two years, from September 2015–September 2017. The need for the second year of technical support from the USGS will be determined by the SFPUC during the first year of model development.

Services will begin immediately on the Effective Date. USGS is expected to provide approximately 150 hours of support over the first year of the Agreement. Technical support will be provided in the form of telephone and remote (email) support. On-site visits may be requested by SFPUC, if needed during model development.

Current versions of software for preparing input data, running the model, and viewing output data will be provided by the USGS.

The following task list provides general guidance to the USGS as to the anticipated scope of work on which USGS may be required to provide support. SFPUC reserves the right to modify this scope of work upon mutual agreement with USGS:

- A.1 Compile climate data from all available climate that include up-to-date climate records.
- A.2 Analyze precipitation and use max and min temperature data and/or PRISM temperature data to calculate temperature lapse rates and correction factors.
- A.3 Generate precipitation correction factors (to consider slope, aspect, and altitude) using PRISM data.
- A.4 Compile soils and geology maps for model domain, incorporate maps into FileGeoDatabase
- A.5. Compile streamflow gages, daily data, and period of records. Add streamflow gages to FileGeodatabase

- B.1. Developing Model Grid and Resample DEM to GRID scale
- B.2. Overlay NHD streams network onto resampled DEM (referred to simple as DEM below)
- B.3. Condition and fill DEM around streams to accommodate required stream resolution
- B.4. Fill DEM using USGS Cascade Routing Tool (CRT), QAQC results of filled DEM
- B.5. Generate stream network using flow direction/flow accumulation
- B.6. Further adjust DEM to assure continuous slopes along streams
- B.7. Create Sub-basins within each watershed with unique IDs

- B.8. Generation of the Streamflow Routing Package Stream Segment and Reach maps
- B.9. Generation of "Clipped" Model Domain and Active Cells Maps, condition DEM along model domain, and generate PRMS HRU IDs
- B.10. Generation of PRMS Gravity Reservoir (GVR) map
- B.11. Add all parameters to HRU and GVR geodatabases

- C.1. Read in HRU and GVR geodatabases and write PRMS parameter file containing the following parameters: • Dimension parameters, Determine initial estimates for `slow_coef_lin` using hydraulic conductivity and HRU slope using kinematic-wave equivalence, Set basin parameters (watershed area, HRU altitudes, etc), Create HRU cascades using CRT cascade generation tool, Define parameters for temperature distribution module (`Temp1sta_prms`), Define parameters for precipitation distribution module (`Precip_prms`), Define parameters for Solar radiation module (`ddsolrad_hru_prms`), Define parameters for potential-evapotranspiration module (`potet_jh_prms`), Define parameters for canopy interception module (`intcp_prms`), Define parameters for snow computation module (`snowcomp_prms`), Compile parameters for surface runoff and infiltration module (`srunoff_smidx_casc`), Compile parameters for soil zone module (`soilzone_gsflow`), Determine initial estimates for `slow_coef_lin` using hydraulic conductivity and HRU slope using kinematic-wave equivalence., Define `ssr2gw_rate` using geology map and recharge potential zonation, Define `sat_threshold`, `soil_moist_max`, `soil_rechr_max`, and `soil_type` using soil maps, Define other parameters using default values and soil maps

- D.1 Define modules and external input file names, define start and end time, output options, and subbasin parameters

- E.1 Evaluate initial output and diagnose
- E.2 Debug parameter file to make successful PRMS test runs
- E.3 Analyze post processing of PRMS Test Runs

- F.1. Calibrate PRMS using measured streamflow as measured variable in objective function.
- F.2. Critically evaluate simulated temperature, precipitation, soil moisture, ET, interflow, runoff, subsurface flow

- G.1. Use data sets developed during steps A-F to construct groundwater a model of watershed that includes streams, ponds, and reservoirs.
- G.2. Compile measured streamflow data, reservoir and pond stages for comparison to GSFLOW simulated values.
- G.3. Adjust model input values to improve comparisons between simulated and measured groundwater heads, streamflow data, and pond levels.
- G.4. Assess model fits to measured data and determine if more formal calibration approaches are required.

2. Deliverables; Records

All deliverables (model input files, reports, etc.) are solely the responsibility of the SFPUC. No deliverable will be provided by USGS to the SFPUC. However, USGS shall maintain accurate documentation and records evidencing services rendered, in the form of written/email correspondence, invoices and receipts, telephone logs, time records, and other data related to the

matters covered by this Agreement. USGS shall make such documentation and records available to the City for review and audit for at least two years after termination of this Agreement.

3. Department Liaison

In performing the services provided for in this Agreement, USGS's liaison with the Department will be **Amod Dhakal, Water Resources Engineer**.

Appendix B

Calculation of Charges

USGS shall perform the requested services on a fixed fee basis, and shall invoice for one lump sum payment annually. The cost for services shall be **\$25,000 per year**. Payment for the first year of services, from the Effective Date to 365 days from the Effective Date, will be made by SFPUC to USGS within 60 days of the Effective Date. SFPUC will assess project progress and determine the need for the second year of services from USGS during the first year of services. Should the Department authorize, in writing, the USGS to perform a second year of services, the Department shall make a one-time payment of **\$25,000 (twenty-five thousand dollars)** to USGS within 60 days of this written authorization. No charges may be incurred by USGS for the second year of services unless approved in writing by the SFPUC Project Manager and SFPUC Bureau Manager.

In no event shall the amount of this Agreement exceed **\$50,000**.